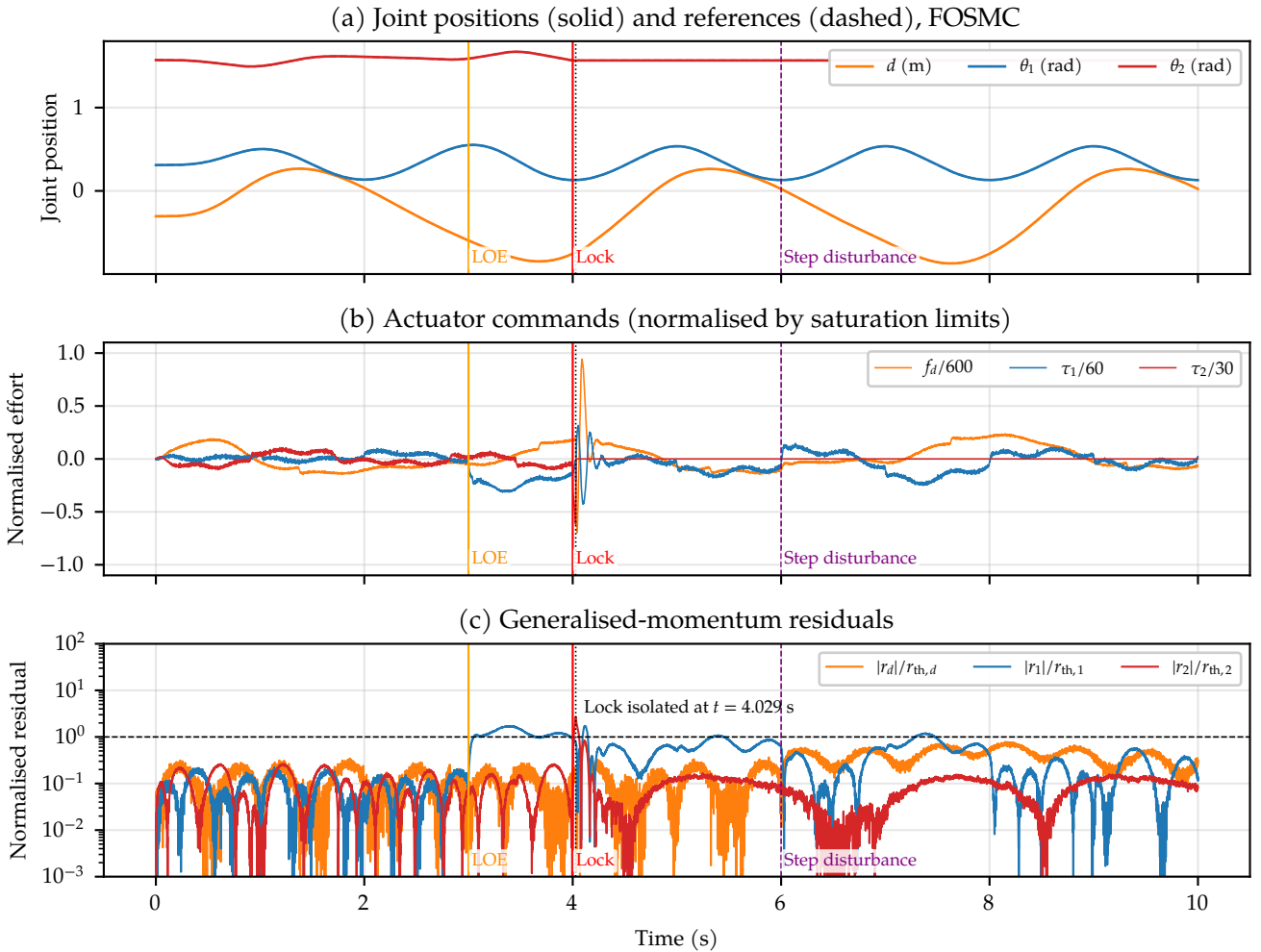

(a) Joint positions (solid) and references (dashed), FOSMC
Joint position
d (m)
θ1 (rad)
θ2 (rad)
LOE
Lock
Step disturbance
(b) Actuator commands (normalised by saturation limits)
Normalised effort
fd/600
τ1/60
τ2/30
LOE
Lock
Step disturbance
(c) Generalised-momentum residuals
Normalised residual
|rd|/rth,d
|r1|/rth,1
|r2|/rth,2
Lock isolated at t = 4.029 s
LOE
Lock
Step disturbance
Time (s)